

NEWCOMER CREDIT DECISIONING COPILOT

Deciding, and defending, thin-file credit for UAE newcomers

A B2B copilot that gives a UAE credit officer a fast, consistent, defensible **approve, decline, or refer** decision on an applicant with no local credit history.

THE PROBLEM

A new arrival starts with a blank credit file

- **The market.** The UAE is about 88% expat, so a large share of loan applicants are recent arrivals.
- **The gap.** A newcomer has no record at the Al Etihad Credit Bureau (AECB), so the usual credit score does not exist.
- **Today.** Banks reject good customers outright, or assess them slowly and inconsistently by hand: the better part of an hour per file.
- **The cost.** Lost good customers, slow onboarding, and decisions an officer cannot easily defend to a regulator.

Five ways a bank solves this today, and the gap each leaves

Player	What it represents	The gap it leaves
AECEB × Nova Credit Passport	Import home-country bureau history	Only covers applicants from networked countries with a usable home file
Newgen Agentic Decisioning	An explainable decisioning engine	Generic engine, not newcomer-specific alt-data scoring
Liv (Emirates NBD)	The incumbent digital bank app	A consumer brand, no explainable thin-file underwriting
Tabby / Tamara (BNPL)	Proof alt-data underwriting works	Small short-tenor, opaque models, not regulator-grade
CRIF Lending Journey	A platform to license vs build	A blank toolkit; the bank still designs the newcomer logic

None owns the whole: newcomer alt-data scoring, a UAE policy check, and a defensible decision for the applicants the others leave invisible. That gap is the product.

The credit officer, who has to stand behind the call

- **User.** A credit officer at a UAE digital bank, assessing newcomer personal-loan and credit-card applications.
- **Their job.** Decide repay-vs-default risk, apply lending policy, and write a rationale they can defend later.
- **Their constraint.** In a regulated decision, being able to explain and reconstruct the call matters as much as the call itself.
- **The design rule.** The tool recommends; the human always makes the final decision.

Rules plus an LLM, with no trained model

- **What we built.** A rules-based scorecard and policy engine for the decision, with an LLM for the explanation layer only.
- **What we did not build.** A trained credit model. No black-box score drives the verdict.
- **Why.** For a regulated v1 lending decision, explainability and defensibility beat a marginal accuracy gain from a model nobody can audit.
- **The featured layer.** The explanation and decision-reasoning, the part a bank cannot get from a raw prompt.

Structured intake to a defensible recommendation

1

Intake

Employment, salary, tenure, rent history, and a few alt-data fields.

2

Scorecard

A transparent 5-factor judgmental scorecard scores repay-vs-default risk.

3

Policy check

Six UAE lending rules run, each with a citation.

4

Explanation

An LLM writes a plain-language rationale grounded only in the above.

5

Recommendation

Approve, decline, or refer. The officer makes the final call.

The score and the verdict are fully deterministic. The LLM writes the explanation only; it cannot change points, rules, or the recommendation.

NOT A CONCEPT, A WORKING PRODUCT

A seven-screen officer console, built in Next.js

- 1 Assess an applicant**
structured intake to a decision: score, rule checks, explanation
- 2 Review queue**
referred cases an officer works; the action persists across a reload
- 3 Audit log**
every decision recorded and reconstructable months later
- 4 Policy**
the live UAE ruleset, read-only, each rule with its citation
- 5 Policy versions**
version history, with each decision stamped by ruleset version
- 6 Policy impact**
champion vs challenger on the 24 profiles, with a live what-if
- 7 Overview**
the dashboard that ties the workflow together

Type-clean and tested. In v1 the decision logic and logs run in the browser; no applicant data leaves the machine. The score and verdict run without any model call.

What the officer sees, end to end

INPUT

Government employee, 14 months in the UAE

Salary AED 20,000 / month, no obligations

On-time rent, 6+ months

Personal loan: AED 40,000 over 12 months

OUTPUT: APPROVE | low risk | 98/100

- ✓ Salary over double the AED 8,000 minimum; debt burden 18% (cap 50%)
- ✓ Government employer, top stability tier
- ✓ All six policy rules pass, each cited
- ✓ 6+ months on-time rent: strongest behavioral signal

A weaker profile returns **refer** for a human to check, or **decline** with the exact rule that failed named. Real output, captured from the running app.

An explanation you can trust, by construction

- **The use case.** The LLM turns the deterministic score and policy results into a plain-language rationale for the officer.
- **Grounding validator.** A check catches any invented fact, retries once, and falls back to a deterministic template, so no ungrounded text reaches the officer.
- **Result.** Hallucination is enforced to zero, not just measured. In the eval, 24 of 24 explanations were grounded on the first attempt.
- **Adversarial probe.** A built-in prompt-injection test proves a malicious input in the name field cannot move the decision.

An exam the model could not study for

- **24 locked profiles.** Synthetic applicants spanning every rule and boundary: 8 approve, 8 decline, 8 refer.
- **Targets set in Phase 1.** The metrics below were our success criteria, defined before the build, not chosen to flatter the result.
- **Labels locked first.** Outcomes were fixed before any scorecard weight existed, so the labels could not be tuned to fit the model.
- **Honesty rule.** Boundary mismatches are surfaced, not tuned away. A mismatch is information about a borderline case, not a bug to hide.

Every safety target met; one latency tail to fix

Metric	Target (set in Phase 1)	Measured	Met
Decision accuracy	80% or higher	20/24 (83.3%)	Yes
False approval rate	under 10%	0/24 (0.0%)	Yes
Hallucination (shown)	0 across the set	0/24	Yes
Policy grounding	100% traceable	100%	Yes
Latency per assessment	under 15s	avg 8.2s, max 15.3s	Tail

Creditworthy newcomers approved: 7 of 8. The four accuracy misses are conservative refers or a cautious call; none is a false approval. The only miss vs target is the latency tail, one slow case at 15.3s.

Each risk has a guardrail and a test, not a hope

The risk	How it is contained and tested
A hallucinated fact in the explanation	Grounding validator catches it, retries, falls back to a template; 0/24 ungrounded, plus an injection probe
A wrong or unsafe approval	24-profile locked evals; 0 false approvals; borderline cases refer, never auto-approve
A decision nobody can defend later	A policy citation on every rule, plus the ruleset version stamped on every decision, in an audit log
A bad policy change shipping	Champion vs challenger run on all 24 profiles before any rule change goes live

The point: a prompt proves the model works once. Production needs evals, override tracking, and decision reconstruction. That is what a bank cannot get from a raw LLM.

Policy is configuration, a market is a pack

- **The rules live in a config pack**, not in code: rule values, citations, score tiers, and band cut-offs are all data.
- **Live what-if.** Drag the debt-burden cap from 50% to 45% and the 24-profile run recomputes on stage, with no code change.
- **The shift is visible.** Approvals drop from 7 of 8 to 6 of 8, two boundary cases move to decline, and every citation now reads 45%.
- **The payoff.** A second market, a new bureau and regulator, is a new pack, not a new project.

From the better part of an hour to minutes

TODAY | manual review

- Assemble salary, visa, obligations, bureau status by hand
- Check each against policy from memory or a PDF
- Write up the rationale
- The better part of an hour per file

WITH THE COPILOT

- ✓ Structured intake
- ✓ Policy rules run instantly, with citations
- ✓ Officer reviews the recommendation and counterfactuals
- ✓ Judgment spent only on the refer queue: minutes

Parked on purpose, ready to answer 'and then?'

- **A second market pack.** Saudi Arabia, swapping AECB and CBUAE for SIMAH and SAMA. The real test of 'a market is a pack'. (Rule values illustrative until SAMA-sourced.)
- **Overrides as feedback.** Every queue override is labeled data on where the rules disagree with human judgment, feeding rule tuning.
- **Version vs version.** A second ruleset (uae v1.1) and a version selector, so impact analysis becomes champion vs challenger.
- **Close the latency tail.** The one item still short of target: isolate the slow case and cap it.

Built, evaluated, and defensible

- All four capstone phases complete: scope, design, build, and evaluation.
- The MVP is built, type-clean, tested, and measured against a locked baseline.
- Every safety target met: 0 false approvals, 0 hallucinations, 100% grounding.
- Live deployment is the one remaining step.

github.com/Mxjoshi/newcomer-credit-copilot

Product, scope, and sign-offs: Monika. Engineering pair: Claude Code.